PEDRO HENRIQUE GONCALVES DE PAIVA

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EDUCATION

Minerva University San Francisco, CA

B.S. in Computational Sciences, Concentration in Data Science and Statistics | GPA: 3.68/4.0 *Aug 2023 – May 2027*

- **Relevant coursework:** Machine Learning, Bayesian Statistics, Probability and Statistics, Econometrics, Algorithms and Data Structures, Linear Algebra, and Modeling Complex Systems.
- Global academic rotations across San Francisco, Taipei, Seoul, Buenos Aires, Berlin, and Tokyo.

RESEARCH & ACADEMIC EXPERIENCE

Minerva University Berlin, DE

Senior Capstone — Satellite Data Analysis *Aug 2026 – Present*

- Conducting an ongoing capstone in satellite data analysis, with particular interest in spatial modeling, multimodal data, model generalization, validation, and uncertainty.
- Exploring how remote-sensing information can be combined with other data sources to produce reliable predictions across changing geographic and environmental conditions.

Universidad Torcuato Di Tella (UTDT) Buenos Aires, AR

Research Assistant, Neuroscience Lab *Sep 2025 – May 2026*

- Built reproducible Python and R pipelines to clean, validate, transform, and link **1,300+ behavioral and cognitive records spanning 15 years**.
- Developed the data model, codebook, validation rules, and quality-assurance procedures used to convert fragmented historical records into an analysis-ready research dataset.
- Supported research linking daily affective measures with cognitive performance in attention, memory, and executive-function tasks.

SELECTED DATA SCIENCE & COMPUTATIONAL PROJECTS

Predicting Spotify Skips from Song Intros — *GitHub · Case Study* *Python · PyTorch · Transformers · scikit-learn | Apr 2026*

- Built a multimodal machine-learning pipeline using **92,445 listening events**, combining behavioral and contextual features with audio representations from convolutional networks and Audio Spectrogram Transformers.
- Identified listening context as the strongest predictor of skip behavior by benchmarking multiple model families on an untouched holdout; logistic regression reached **75.6% accuracy and 0.822 ROC-AUC**.
- Analyzed what the performance gap revealed about feature representation, model complexity, and the information actually available for prediction.

Dengue Surveillance Dashboard — *GitHub · Live* *Python · pandas · scikit-learn · statsmodels · Streamlit · Plotly | Apr 2026*

- Built and deployed an end-to-end public-health data product covering **five Brazilian states**, transforming official surveillance records into weekly forecasts, risk indicators, uncertainty intervals, and interactive visualizations.
- Compared forecasting approaches using leakage-safe, expanding-window evaluation and improved pipeline scalability through **250,000-row chunk processing**, Parquet caching, automated tests, and continuous integration.

Bayesian Modeling of Football Attendance — *GitHub* *Python · PyMC · ArviZ · Bayesian Statistics · Jupyter | Dec 2025*

- Developed a hierarchical Negative Binomial model of attendance across **12 Argentine football clubs**, incorporating team-level and weekday effects.
- Evaluated model fit and uncertainty through MCMC diagnostics, posterior predictive checks, and PSIS-LOO; hierarchical modeling improved out-of-sample predictive performance by **18.4 ELPD points** over complete pooling.

Wildfire Spread on a Real Deforestation Frontier — *Report · GitHub* *Python · Simulation · Cellular Automata | Dec 2025*

- Developed a cellular-automata simulation using real Amazon deforestation patterns to study how spatial intervention strategies alter simulated wildfire propagation.
- In simulation, targeted intervention strategies reduced burned area by up to **83.8% across 500 Monte Carlo runs**.

Additional computational projects

- Developed a constraint-aware scheduler using a custom max-heap and dependency handling; reconstructed genealogical relationships from genetic sequences using dynamic programming, pairwise similarity matrices, and greedy/global optimization approaches.

LEADERSHIP EXPERIENCE

Yale University, Yale Young Global Scholars — New Haven, CT

Instructor & Capstone Supervisor, Innovations in Science & Technology — *Jun 2026 – Aug 2026*

- Built and evaluated a **retrieval/RAG workflow** for unstructured text using TF-IDF, cosine similarity, top-k retrieval, and an LLM.
- Designed and taught seminars on recommendation systems, algorithmic bias, and the ways algorithmic systems shape information and behavior to **135 international high-school students**.
- Supervised student capstone work and supported the development of interdisciplinary technical projects.

Minerva University — San Francisco, CA

Teaching Assistant, Introduction to Computer Science — *Aug 2024 – May 2026*

- Supported **200+ students** in programming, quantitative reasoning, statistics, and formal logic.
- Diagnosed conceptual and implementation errors during coursework and adapted technical explanations to students with different levels of prior computational experience.

AURA Application Mentorship — Remote, Brazil

Co-Founder & Product Lead — *Jan 2023 – Present*

- Analyzed engagement, application outcomes, and student feedback across **400+ students and 14 cohorts** to identify participation gaps and improve program design.
- Co-developed curriculum, digital resources, and advising systems that have supported **700+ university acceptances worldwide**.

REC — Cornerstone Civic Project — San Francisco, CA

Product Strategy & Analytics Lead — *Aug 2023 – Apr 2024*

- Conducted audience and stakeholder research to identify user needs, potential partners, and barriers to engagement, translating findings into product priorities and measurable outcomes.
- Built a structured feedback-analysis workflow in Google Sheets to synthesize user responses, identify recurring themes, and guide iterative improvements to the project.

HONORS & AWARDS

1st Place, OPTIC Intercollegiate Forecasting Tournament — Ranked first in a multi-university probabilistic forecasting competition evaluating accuracy and calibration across real-world questions.

Best Hack for Quality of Life & Best Use of the TwelveLabs API, UC Berkeley Hack for Impact — Developed *RememberMe*, an AI-assisted memory-support prototype.

Winner, AI Consensus, Student Essay Contest on AI and Education — Selected for publication by *The Centaur* and awarded a $200 writer stipend.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, pandas, NumPy, SciPy, Jupyter.

ML & Statistics: PyTorch, scikit-learn, PyMC, ArviZ, statsmodels; Bayesian modeling, regression/classification, forecasting, model validation.

Methods: time-series modeling, Monte Carlo simulation, data pipelines, dynamic programming.

Languages: Portuguese (native), English (fluent), Spanish (intermediate).